Keep the Angle

A Geometry-Preserving Basis in Spectral Embeddings

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Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [14]. We show the geometry does not vanish; it migrates entirely into the *angular* coordinate. Writing each node's low-mode embedding in polar form (magnitude × direction), the direction preserves graph geodesics at Spearman $\rho \approx 0.93$ on a reference manifold, while the magnitude preserves $\rho \approx 0.03$ and a random-mode basis of equal size preserves $\rho \approx 0$. Keeping only the angle is exactly the row-normalization of Ng–Jordan–Weiss spectral clustering [8], whose existing theory concerns cluster separation [12]; we give it a geodesic justification. We show, as a corollary of von Luxburg–Radl–Hein, that the radial coordinate of the full embedding degenerates to a local-degree quantity for intrinsic dimension $d \geq 2$, and **conjecture**, with strong empirical support, that the angular coordinate remains rank-faithful to geodesic distance—flat in both mode-count and graph size where the raw embedding decays. The basis is **universal**: it preserves geodesics on any substrate carrying genuine low-dimensional Riemannian geometry—random-geometric lattices, manifold-embedding trajectories, and Wolfram-model hypergraph rewriting alike—and fails correctly on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ($d \approx 1$–3.6), fidelity falls approximately linearly with intrinsic dimension. Refined toward the continuum (N up to 5×10^4), the basis *sharpens* on sampled Riemannian substrates (angular-fidelity Spearman $+1.00$ with $\log N$ at d=3) yet *decays* below a pre-registered floor on a coordinate-free Wolfram-model manifold—a dissociation that recasts the angular coordinate as a *diagnostic* for whether a substrate's fine structure stays manifold-like under refinement.

1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [5]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [14]. Yet the most successful spectral-clustering algorithm [8] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [12, 13]. It does not address metric geometry. **This paper asks what row-normalization recovers of the graph's geodesic structure, and why.**

Our answer is a clean decomposition. Write each node's low-mode embedding in polar form, magnitude × direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg's theorem kills; the *direction* carries the graph's geodesic geometry. Row-normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

This "keep the scale-invariant direction" operation recurs beyond clustering—for example in polar-coordinate KV-cache quantization, which transfers the scale-invariant direction and quantizes the radius separately [10]. It also admits a physical reading: in observer theory a bounded observer perceives geometry only after coarse-graining a discrete substrate [17], and the angular projection is a candidate for that coarse-graining. We keep that interpretation to the discussion (§7) and let the spectral result stand on its own; the empirical core is that the basis is *universal*, preserving geodesics on any graph carrying genuine low-dimensional Riemannian geometry, independent of how it was generated—and, we show in §5.8, the same angular diagnostic reads out on a real semantic embedding space, a language model's moral-text vectors.

Contributions: (1) the **Keep-the-Angle Principle** (§3)—a Radial-Degeneracy *corollary* (for the full embedding, from [14]) with a spectral-gap bound on the truncation error (Proposition 1) and an empirically-supported Angular-Preservation *conjecture* now with first metric-distortion evidence; (2) a measurement with controls (§5); (3) universality across five substrate types with a two-factor small-world screen (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); (5) a cross-domain probe on a language-model semantic embedding (§5.8); and (6) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings.

2 Background

Spectral geometry. Laplacian eigenmaps [1] and diffusion maps [5] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commute-time degeneracy.** von Luxburg et al. [14] proved the commute distance of a large geometric graph degenerates, $R(i,j) \to 1/k_i + 1/k_j$ (k_i the degree of node i); we read this backwards as the collapse of the *radial* coordinate to local degree. **Row-normalization.** Ng–Jordan–Weiss [8] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [12] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions, explaining k-means' post-normalization success; consistency of spectral clustering is treated in [13]. Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. **Emergent geometry and observer theory.** In the Wolfram model [15, 16, 6] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [17]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [7, 11, 9]; we test and reject a hyperbolic reframe in §6.

3 The Keep-the-Angle Principle

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k), $0 = \lambda_0 < \lambda_1 \leq \cdots$. The low-mode spectral embedding using the lowest m nontrivial modes places node i at $\Psi_i = \left(v_1(i)/\sqrt{\lambda_1}, \ldots, v_m(i)/\sqrt{\lambda_m}\right)$. Write $\Psi_i = r_i\,\hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$. [1]

[1] We call Ψ "commute-style" rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all $n-1$ modes, so that $\|\Psi_i - \Psi_j\|^2$ equals the resistance

Intrinsic dimension and degree notation. Throughout, d is the intrinsic dimension of the substrate, and k_i denotes the degree of node i (we reserve d for dimension to avoid the collision d vs. d_i): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices, sphere), and for emergent graphs the value at which the three independent estimators of §4 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. The regime $d \geq 2$ refers to this measured intrinsic dimension; §5.4 confirms the $d = 1$ boundary behaves as predicted.

Corollary 1 (Radial Degeneracy). *Let G_n be random geometric graphs of intrinsic dimension $d \geq 2$ satisfying the conditions of [14], and let Φ_i be the* full *commute-time embedding built from the unnormalized Laplacian, so that $\|\Phi_i\|^2 = L^+_{ii}$. As $n \to \infty$ the resistance distance degenerates, $R(i,j) \to 1/k_i + 1/k_j$; since the only per-node content of this additively separable form is the self-term L^+_{ii}, the radius satisfies $\|\Phi_i\| \to 1/\sqrt{k_i}$. The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density coordinate carrying no geodesic information.*

Remark (Truncated, normalized radius). Our experiments use the truncated radius $r_i = \|\Psi_i\|$ (m lowest modes of L_{sym}), not $\|\Phi_i\|$. Transferring Corollary 1 to this setting needs (i) the normalized analogue of the degeneracy [14] and (ii) control of the truncation error. We settle (ii) unconditionally below and support (i) numerically; the residual gap is then the angular claim itself (Conjecture 1). Numerically the degeneracy holds tightly: $\rho(r_i, 1/\sqrt{k_i}) = 0.92$–$0.99$ and radius-only geodesic preservation $\rho \leq 0.08$ (§5.5).

Proposition 1 (Truncation is spectral-gap bounded). *Let $\Psi^\infty_i = \big(v_k(i)/\sqrt{\lambda_k}\big)^{n-1}_{k=1}$ be the full symmetric-normalized commute-style embedding and Ψ^m_i its truncation to the lowest m nontrivial modes. Then for all nodes $i \neq j$,*

$$0 \leq \|\Psi^\infty_i\|^2 - \|\Psi^m_i\|^2 \leq \frac{1}{\lambda_{m+1}}, \qquad 0 \leq \|\Psi^\infty_i - \Psi^\infty_j\|^2 - \|\Psi^m_i - \Psi^m_j\|^2 \leq \frac{2}{\lambda_{m+1}}.$$

Proof. Each left inequality is termwise: truncation deletes only the summands with $k > m$, of the form $v_k(i)^2/\lambda_k \geq 0$ or $(v_k(i) - v_k(j))^2/\lambda_k \geq 0$. For the right inequalities, $\lambda_k \geq \lambda_{m+1}$ for every $k > m$, so the deleted mass is at most $\lambda^{-1}_{m+1}\sum_{k>m} v_k(i)^2$ and $\lambda^{-1}_{m+1}\sum_{k>m}(v_k(i) - v_k(j))^2$ respectively. The eigenvectors $\{v_k\}^{n-1}_{k=0}$ are orthonormal, so $VV^\top = I$ gives $\sum_k v_k(i)^2 = 1$ and, for $i \neq j$, $\sum_k v_k(i)v_k(j) = 0$; hence $\sum_{k>m} v_k(i)^2 \leq 1$ and $\sum_{k>m}(v_k(i)-v_k(j))^2 \leq \sum_k (v_k(i)-v_k(j))^2 = \sum_k v_k(i)^2 + \sum_k v_k(j)^2 - 2\sum_k v_k(i)v_k(j) = 2$. ☐

Two consequences. First, the truncation error is *bounded independently of n*: it is controlled entirely by the inverse spectral gap $1/\lambda_{m+1}$, so refining the graph at fixed m cannot amplify it—the mechanism behind the continuum survival of §5.7. Second, the deleted content is high-frequency (large-λ); the geometry lives in the retained low modes (an equal-size random-mode basis preserves $\rho \approx 0$, §5), so the bound formalizes why aggressive truncation does not lose the geometry—it discards a spectral-gap-bounded, high-λ remainder. On the 2-torus ($n = 2000$) the bound holds with room to spare: the realized radius error is 3.5–17% of $1/\lambda_{m+1}$ across $m \in \{5, 10, 20, 40\}$, the full normalized radius is near-constant (coefficient of variation 0.09), and the truncated radius stays rank-correlated with $1/\sqrt{k_i}$. What remains open is not the radius but the angle: Proposition 1 bounds how far truncation moves the metric, not that the surviving *angular* coordinate equals the geodesic—that is Conjecture 1.

$R(i,j)$ exactly. Our Ψ is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Corollary 1 and is handled there.

Conjecture 1 (Angular Preservation). *Row-normalization $\hat{u}_i = \Psi_i/\|\Psi_i\|$ deletes the degenerate density factor of Corollary 1, and the pairwise distances among the resulting directions remain* rank-faithful *to geodesic distance—Spearman correlation bounded away from zero (empirically ≈ 0.9)—* uniformly in *the mode-count m and size n. A stronger, metric form of the conjecture—that the angular map is bi-Lipschitz to geodesic distance with constants uniform in m and n—is consistent with our data but untested; our evidence is rank-based and cannot by itself bound metric distortion.*

Metric evidence on the torus. The metric form is no longer entirely untested. On the 2-torus we compare embedded distance to graph geodesic directly. After a single global scale, the angular map has a robust empirical bi-Lipschitz constant $\kappa \approx 2.6$ (ratio of the 97.5th to 2.5th percentile of $d_{\text{emb}}/d_{\text{geo}}$ over 19,900 pairs), tighter than the full commute embedding (3.1), an equal-size random-mode basis (8.3), or the magnitude coordinate (125); and its leading four coordinates Procrustes-match the canonical flat-torus embedding in $\mathbb{R}^4$ at disparity 0.07, versus 0.995 for random modes. This is genuine metric distortion, not rank correlation—one substrate, but a well-understood one. (The full-range constant is inflated by the few near-antipodal pairs the torus folds together, hence the percentile-robust κ.) A uniform-in-(m, n) bi-Lipschitz bound remains the open metric conjecture; this is one datapoint toward it.
Status. Empirically strong (§5) but **not proven**. It rests on eigenmap-embedding results plus a heuristic; the honest gap is that Corollary 1 explains why the radius *dies* for $d \geq 2$, not by itself why the angle *lives*—indeed the angular fidelity persists ($\rho \approx 0.88$–0.90) even on quasi-1D graphs where Corollary 1 does not apply, so the angular half is carried by the eigenmap-embedding mechanism, not degeneracy alone. That mechanism is analyzable rather than mysterious: if the low L_{sym} eigenvectors converge to Laplace–Beltrami eigenfunctions [2, 3] and eigenfunction coordinates furnish locally bi-Lipschitz charts of the underlying manifold [4], the angular map's geodesic fidelity should follow from the chart geometry—the natural route to a proof of Conjecture 1, and this paper's central open problem. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i.

Dimension gating. Corollary 1 is a $d \geq 2$ phenomenon. In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Corollary 1 correctly does not apply—a prediction, not a failure, confirmed in §5.5.

4 Methods

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread $\Rightarrow$ clean manifold, and the agreed value is our measured intrinsic dimension d. **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (unit-normalized low-mode embedding), *random* (m random modes; control), *magnitude* (radius only), *euclid* (classical 2D MDS). **Uncertainty.** Each ρ is a rank correlation over $\binom{n_{\text{anchor}}}{2}$ node pairs ($n_{\text{anchor}} = 150$ throughout, 200 for the cross-domain probe of §5.8—11,175 and 19,900 pairs respectively). Bootstrap 95% confidence intervals over pair resampling are tight ($\leq \pm 0.005$ on the headline numbers); the dominant source of variation is the graph draw itself, so for headline ρ we also report spread over independent draws (2-torus, m=10: 0.914 ± 0.009 over five draws). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding trajectories. The swiss-roll case is a synthetic 2D manifold—intrinsic coordinates (u, v), u the roll angle and v the height—carried by a random orthogonal map

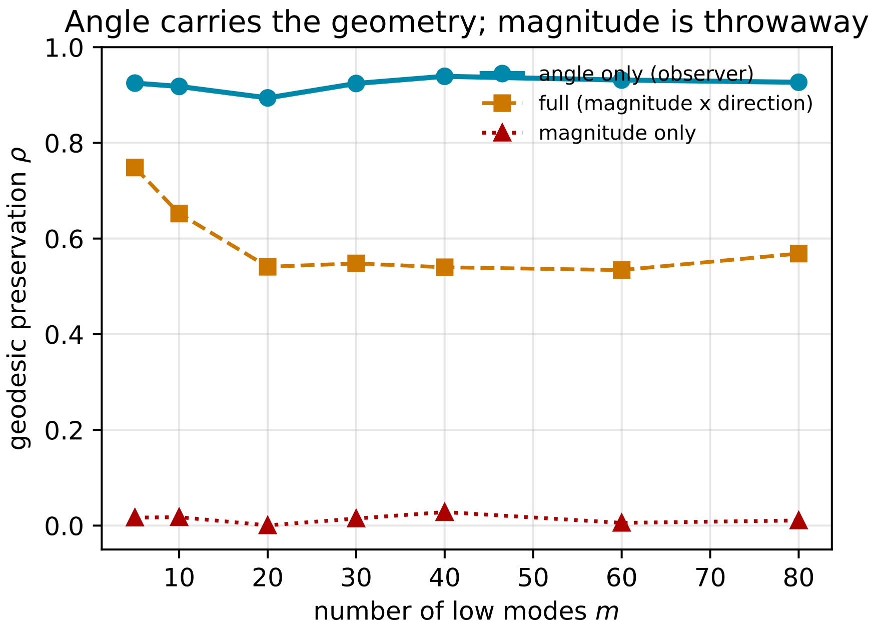

Figure 1: Spearman correlation with graph geodesics on a 2-torus ($n = 2000$) as a function of mode-count m. Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with m.

into $\mathbb{R}^{12}$ with isotropic noise (σ=0.03) and connected by a k=10 nearest-neighbour graph on n=2000 ambient points. It stands in for sequential/embedding-derived data (each item a point on a low-d semantic manifold, consecutive items nearby), not a specific text corpus.

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3. On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag.×dir.)	**angle only**	magnitude only
5	0.761	**0.926**	0.025
10	0.650	**0.913**	0.035
20	0.550	**0.891**	0.028
40	0.517	**0.928**	0.034

The angle carries the geometry (≈ 0.92, flat in m); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2).

5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only $\approx$ 1.5D manifolds (82 manifold-grade rules in a 20k sweep; none at $d \geq 2.5$). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020

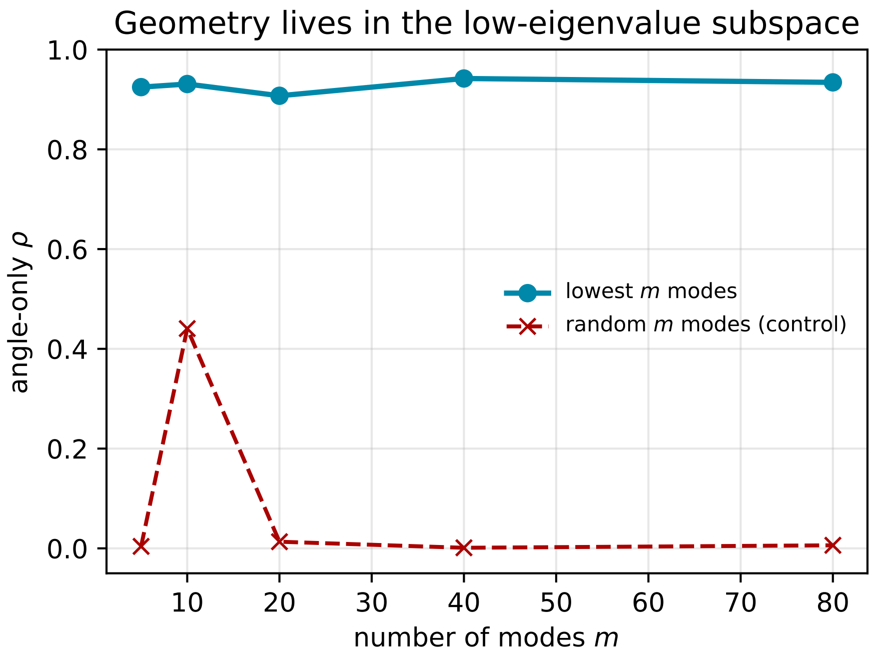

Figure 2: Geometry lives in the low-eigenvalue subspace: lowest-m modes preserve geodesics; a random-mode basis of equal size does not.

rule ($d = 2.07 \pm 0.01$, spread 0.05, angle-$\rho = 0.82$) and by from-scratch evolutionary search (≈ 3 generations). The angular basis holds on genuine 2D emergent space.

5.3 Universality across substrates

substrate	meas. d	angle-ρ	random	euclid
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll trajectory	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	3.9	0.57	0.01	0.55
small-world (destroyed)	3.0*	0.45	0.00	—

Substrates with genuine low-d Riemannian geometry all show the effect, independent of graph origin; the metric fails correctly where geometry is Lorentzian or destroyed. *For the small-world graph the three dimension estimators disagree sharply (ball 3.7, spectral 3.6, effective-rank 4.3; spread 0.77 versus ≤ 0.14 on the clean substrates); by our own manifold-grade criterion this graph is *not* a manifold, and the low angle-ρ is the predicted failure, not an exception. The two-factor screen below makes this quantitative.

Two caveats keep the universality claim honest. First, on the two flat substrates (king-lattice, swiss-roll) classical MDS *beats* the angular basis (0.95 and 0.99 vs. 0.82 and 0.93): MDS wins whenever the manifold embeds near-isometrically in low-dimensional Euclidean space, while the angle wins where topology obstructs such an embedding (the torus: 0.89 vs. 0.67) and, unlike MDS, is uniform in m and n (§5.5). Universality here means the angular basis never *loses* the geometry on a genuine low-d Riemannian substrate—not that it dominates every baseline on every substrate. Second, the swiss-roll trajectory is the sequential-embedding case: the angular basis is the same whether the manifold comes from hypergraph rewriting or sequential trajectory data.

A two-factor small-world screen. To make "destroyed" precise rather than a label, we screen every substrate on two graph statistics against degree-matched baselines—average clustering C (versus an Erdős–Rényi C_{rand} and a ring-lattice C_{latt}) and characteristic path length L (versus L_{rand})—combined into the Telesford index $\omega = L_{\text{rand}}/L - C/C_{\text{latt}}$ ($\omega < 0$ lattice-like, $\omega \approx 0$ small-world, $\omega \to 1$ random). Clean low-d manifolds sit firmly on the lattice side; the tangles do not:

substrate	angle-ρ	ω	dim-spread	verdict
2-torus	0.91	-0.55	0.14	clean
king-lattice	0.83	-0.50	0.51	clean
swiss-roll	0.93	-0.66	0.45	clean
king-lat. rewired	0.42	$+0.23$	0.77	small-world
causal set	0.57	$+0.75$	7.87	non-manifold

The screen robustly separates the genuine manifolds ($\omega \leq -0.50$) from strongly destroyed geometry ($\omega \geq +0.24$), and the inter-estimator dimension spread corroborates (≤ 0.51 clean versus ≥ 0.77 contaminated): the low angle-ρ on the rewired and Lorentzian substrates is small-world/non-manifold contamination, not a failure of the basis on genuine geometry. A king-lattice rewiring sweep both sharpens and honestly bounds this. As the rewire probability rises, angle-ρ collapses almost at once ($0.81 \to 0.47$ by 1% rewiring) while ω does not cross zero until $\approx 5\%$; so $\omega < 0$ is a *conservative* flag for destruction, and in the transition band (1–3%) angular fidelity is in fact the *more* sensitive detector of geometric corruption than either clustering or dimension spread. We therefore read the screen as confirming gross contamination, with angle-ρ itself the finer instrument—the mechanism behind the scaling-law outlier below.

5.4 The scaling law

Across 20 independent emergent manifolds ($d \approx 1$–3.6),

$$\text{angle-}\rho \;\approx\; 1.09 - 0.157\,d, \qquad \text{slope } -0.157 \pm 0.028, \quad R^2 = 0.64, \quad d \in [1.0, 3.6].$$

(Spearman -0.881 [95% CI $-0.97, -0.67$; $n = 20$ manifolds] between ρ and d is a distribution-free monotonicity check; the linear fit's slope has a bootstrap 95% CI of $[-0.185, -0.132]$, comfortably excluding zero. The modest $R^2 = 0.64$ reflects a single outlier—a $d{\approx}1.97$ manifold at $\rho{=}0.39$—which the small-world screen of §5.3 flags as shortcut-contaminated rather than clean geometry.) The fit is *empirical and local*: ρ is a rank correlation bounded by 1, so the linear form holds only within the measured range—the intercept 1.09 is an extrapolation, not a literal $d \to 0$ fidelity, and the slope may vary with substrate roughness. A bounded model fits essentially as well and respects $0 < \rho < 1$: the logistic $\rho \approx \left[1 + e^{0.82\,(d-3.66)}\right]^{-1}$ gives $\rho(0) = 0.95$ and $\rho \to 0$ at $R^2 = 0.62$, against the linear 0.64. Exponential forms do not help—an unoffset Ae^{-kd} still has $A = 1.18 > 1$ and an offset version drives the asymptote negative. Within range the two are visually indistinguishable; we keep the linear coefficient as the local slope and note the logistic as the boundary-respecting global form. Within range, fidelity is highest at low dimension and softens toward ≈ 0.56 at $d \approx 3.4$ (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

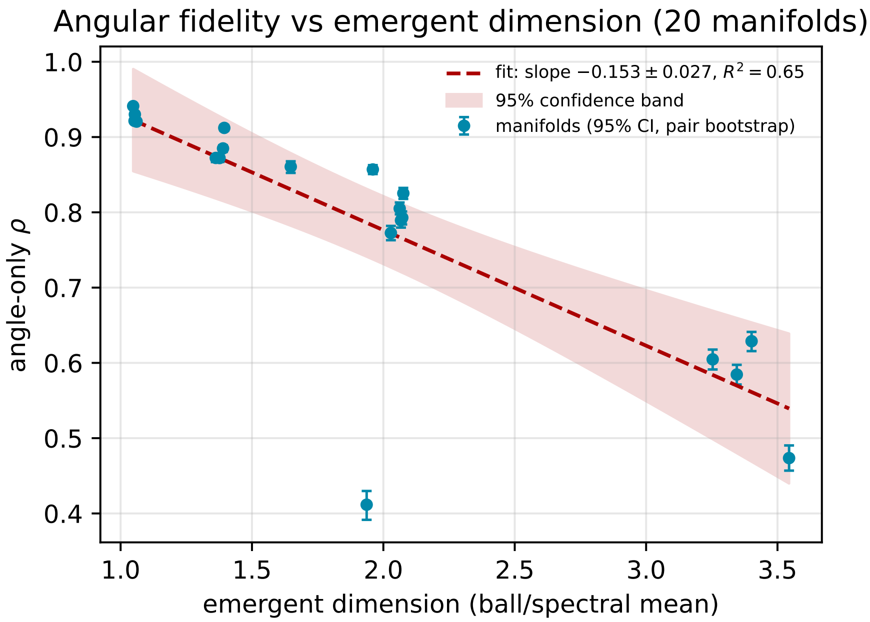

Figure 3: Angular fidelity falls linearly with emergent dimension across ring, tori, and Wolfram-model rules; the shaded band is the 95% bootstrap CI on the slope ($[-0.185, -0.132]$). Per-point pair-resampling CIs are smaller than the plot markers ($\leq \pm 0.005$).

5.5 Verifying Corollary 1 and Conjecture 1

angle-ρ is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ($\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.92$–$0.99$; radius-only $\rho \leq 0.08$). Corollary 1 holds cleanly at $d = 3$ ($\rho(R, 1/k_i + 1/k_j) \to 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 1: in an m-sweep at $n = 4000$, angle-ρ is flat ≈ 0.93 across $m = 5$–80 while the full embedding decays $0.773 \to 0.437$.

5.6 Stability across substrate evolution

Universality across substrate *types* (§5) is a statement about fixed graphs. The angular basis also persists as a substrate *evolves*. For each of 151 qualifying Wolfram-model rules (95 arity-2, 56 arity-3) we embed successive rewrite snapshots on a fixed anchor set and compare, between consecutive snapshots, the Spearman persistence of the *angular* pairwise structure (`consec`) against the persistence of the underlying *geodesic* structure itself (`churn`)—the substrate's own rate of change is the baseline the basis must keep pace with. On all 143 rules resolvable at protocol scale (8 never yield ≥ 3 distinct snapshots and are excluded from the denominator but reported), the angular structure tracks the evolving geometry at least as stably as the graph's own geodesics, $\overline{\texttt{consec}} \geq 0.9\,\overline{\texttt{churn}}$: a $143/143 = 100\%$ pass against a pre-registered 80% bar, with zero substrate-unstable rules. Over the longest time-lever (first vs. last snapshot) angular persistence declines only gently with emergent dimension,

$$\text{slope } -0.029/\text{dim}, \quad \text{Spearman } -0.48, \quad d \in [0.98, 2.23],$$

and this replicates independently in both arities (arity-2 -0.028/dim, $\rho = -0.45$, $n = 95$; arity-3 -0.029/dim, $\rho = -0.54$, $n = 48$)—the same mechanism, unchanged by moving from binary to ternary rewriting.[2]

5.7 Continuum survival, and a dissociation

Every result so far is at finite N ($\approx$ 2–4k nodes, capped by dense `eigh`); the "uniform in N" clause of Corollary 1 and Conjecture 1 is untested at scale (§5.5). We test it directly by refining each substrate toward the continuum with a sparse eigensolver, $N \in \{2, 5, 10, 20, 50\} \times 10^3$, four seeds, $m = 10$ modes, $n = 300$ anchors—well past the $\sim$ 5k dense ceiling. The two substrate families dissociate. On genuine Riemannian substrates the basis *sharpens*: on random geometric graphs the angular fidelity is flat-to-rising in N, most cleanly in the higher-dimensional case,

$$\begin{aligned} \text{RGG } d{=}3: \quad & \text{angle-}\rho: \ 0.856 \to 0.912 \ \ (N: \ 2\text{k} \to 50\text{k}), \\ & \text{Spearman}(\text{angle-}\rho, \log N) = +1.00, \end{aligned}$$

with $d{=}2$ likewise monotone (+0.90, 0.916 $\to$ 0.929): refining the mesh makes the low modes resolve the geometry *better*, not worse, and both controls stay pinned (higher-mode $\rho \le 0.05$, magnitude $\rho \le 0.03$, eigensolver residual $\sim 10^{-14}$) at every N. The finite-N evidence of §§5–5.5 was not a small-graph artifact.

The dissociation is with *emergent* substrates. The $\mathrm{R_{3D}}$ Wolfram-model manifold—geometry with no latent coordinates—*decays* monotonically with refinement, angle-ρ : 0.749 $\to$ 0.654 (Spearman = -1.00), falling below a pre-registered floor of 0.75; and at the largest N its magnitude control begins to break the clean angle/radius separation that holds exactly on the RGGs (one 50k seed crosses $\rho_{\mathrm{mag}} = 0.33$ and is gated out). A sampled manifold refines to a clean continuum; an emergent hypergraph, under this basis, does not—its fine structure drifts away from a manifold as it grows. The coordinate-free substrate—the one the observer-theory reading most needs—is exactly the one where the basis fails in the limit. We read this not as a limitation but as a result: the angular basis is a *diagnostic* for whether a substrate's fine structure stays manifold-like under refinement, and it returns different verdicts for sampled (2/3 pass) and emergent (1/1 fail) geometry.[3]

5.8 A cross-domain probe: the moral embedding manifold

If the angular basis is a diagnostic for manifold-like structure (§5.7), it should return an interpretable reading on a substrate from an entirely different domain. We point the same instrument at a *semantic* space: 2,400 moral/ethical text items embedded with BGE-M3 (1024-d), stratified across six traditions, connected by a k=10 nearest-neighbour graph on the unit-normalized embeddings. The moral embedding reads as substantially manifold-like—angle-ρ = 0.893 over 19,900 pairs, against a per-feature-shuffle control at 0.233 and an isotropic-Gaussian control at 0.311, with both mode controls pinned (higher-mode $\rho = 0.01$, random-mode $\rho = 0.07$). The same basis that reads 0.93 on a swiss roll and ~ 0 on a tangle places moral-semantic space near the manifold end. But the diagnostic also localizes where it departs from a textbook Riemannian substrate: the three dimension estimators disagree (spread 2.6; effective-rank 3.8 against spectral 1.2), the magnitude coordinate carries real signal ($\rho_{\mathrm{mag}} = 0.56$, versus ≤ 0.05 on the clean geometric substrates), and

[2] Pre-registered as Amendment 2: an earlier capture hook could append the same state once per crossed edge-count threshold, inflating consecutive stability toward a spurious 1.0. The corrected rule takes at most one snapshot per generation with dynamic 2× spacing; the numbers here are from that harness.

[3] Pre-registered (PREREG_RUNG4) before the run; survival bar angle-$\rho(N_{\max}) \ge \max(0.75, 0.9\,\text{angle-}\rho(N_{\min}))$ on $\ge 2/3$ substrates. Amendment 1, registered before any run: the whole-spectrum random-mode control needs a dense decomposition unavailable at $N \ge 50$k, so it is the top m of the lowest-64 sparse-computed modes—a faithful, sparse-affordable stand-in, same ≤ 0.15 bar.

the small-world screen returns $\omega = +0.15$—a clustered space with mild shortcut structure and a nontrivial radial mode, not a pure low-d manifold. The angular coordinate transfers across domains as a graded diagnostic, not merely a pass/fail label.

6 Honest Negatives

Not secretly hyperbolic (rejected). With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw -0.98 against update density that deflates to a tautology (update-count equals degree, $\rho = +1.00$; hubs are negatively curved); controlling for degree, the independent signal is partial-$\rho = -0.14$. This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

7 Discussion and Conclusion

The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a geodesic justification complementary to the cluster-separation theory of [12], unifies it with the scale-invariant direction of KV-cache compression [10], and holds *universally* across substrates with genuine low-d Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-rewriting and sequential-trajectory substrates is evidence that scale-invariant angular coarse-graining is a general mechanism by which relational data carries geometric structure.

One interpretation is observer-theoretic. In Wolfram's observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making "the observer's basis" a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy- responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving Conjecture 1 (in its rank form, and ideally the bi-Lipschitz strengthening); extending Corollary 1 to the truncated, normalized radius actually used in practice; synthesizing clean manifolds above $d \approx 2$; and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude: the basis-level fact is the contribution; the observer interpretation is where we leave the door open.

References

[1] M. Belkin and P. Niyogi. Laplacian eigenmaps for dimensionality reduction and data representation. *Neural Computation*, 15(6):1373–1396, 2003.

[2] M. Belkin and P. Niyogi. Convergence of Laplacian eigenmaps. In *Advances in Neural Information Processing Systems 19*, 2007.

[3] N. García Trillos and D. Slepčev. A variational approach to the consistency of spectral clustering. *Applied and Computational Harmonic Analysis*, 45(2):239–281, 2018.

[4] P. W. Jones, M. Maggioni, and R. Schul. Universal local parametrizations via heat kernels and eigenfunctions of the Laplacian. *Annales Academiæ Scientiarum Fennicæ Mathematica*, 35:131–174, 2010.

[5] R. Coifman and S. Lafon. Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1):5–30, 2006.

[6] J. Gorard. Some relativistic and gravitational properties of the Wolfram model. *Complex Systems*, 29(2):107–192, 2020.

[7] M. Gromov. Hyperbolic groups. In *Essays in Group Theory*, MSRI Publications 8, pp. 75–263, Springer, 1987.

[8] A. Ng, M. Jordan, and Y. Weiss. On spectral clustering: analysis and an algorithm. *Advances in Neural Information Processing Systems 14 (NeurIPS)*, 2002.

[9] M. Nickel and D. Kiela. Poincaré embeddings for learning hierarchical representations. *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.

[10] I. Han, P. Kacham, A. Karbasi, V. Mirrokni, and A. Zandieh. PolarQuant: Quantizing KV caches with polar transformation. *arXiv preprint arXiv:2502.02617*, 2025. `https://arxiv.org/abs/2502.02617`. Accessed July 2026.

[11] R. Sarkar. Low-distortion delta-hyperbolic embedding in the hyperbolic plane. *Graph Drawing*, LNCS 7034, pp. 355–366, Springer, 2011.

[12] G. Schiebinger, M. J. Wainwright, and B. Yu. The geometry of kernelized spectral clustering. *Annals of Statistics*, 43(2):819–846, 2015.

[13] U. von Luxburg, M. Belkin, and O. Bousquet. Consistency of spectral clustering. *Annals of Statistics*, 36(2):555–586, 2008.

[14] U. von Luxburg, A. Radl, and M. Hein. Hitting and commute times in large random neighborhood graphs. *JMLR*, 15:1751–1798, 2014.

[15] S. Wolfram. *A New Kind of Science*. Wolfram Media, 2002.

[16] S. Wolfram. Finally we may have a path to the fundamental theory of physics... and it's beautiful. Writings blog, April 2020. `https://writings.stephenwolfram.com/2020/04/finally-we-may-have-a-path-to-the-fundamental-theory-of-physics-and-its-beautiful/`. Accessed July 2026.

[17] S. Wolfram. The concept of the ruliad. Stephen Wolfram Writings blog, November 2021. `https://writings.stephenwolfram.com/2021/11/the-concept-of-the-ruliad/`. Accessed July 2026.